

IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 21

Create Security Team Permission Sets and Persona-Based Access

IAM Identity Center | Permission Sets | Security Personas

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ENVIRONMENT	Personal AWS Lab
VERSION	v1.0.0
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STATUS	Complete

Executive Summary

Expanded IAM Identity Center access into a persona-based access model. Added **SecurityAudit**, **IAM**, **SecurityOperations**, and **TestAccount1** to the existing **AdministratorAccess** path, transitioned administration into the delegated **IAM** account, created **Security Administrators** and **IAM Administrators** groups, created **ReadOnlyAccess**, **IdentityCenterAdmin**, and **SecurityFullAdmin** permission sets, then assigned those permission sets to the appropriate accounts and groups.

Business Problem

Using full administrator access for every task is risky, even in a lab. A security team needs different personas: a read-only role for investigation, an identity administration role for IAM Identity Center operations, and a full-admin role for break-glass security response. IAM Identity Center permission sets provide that separation while still using centralized SSO.

Objectives

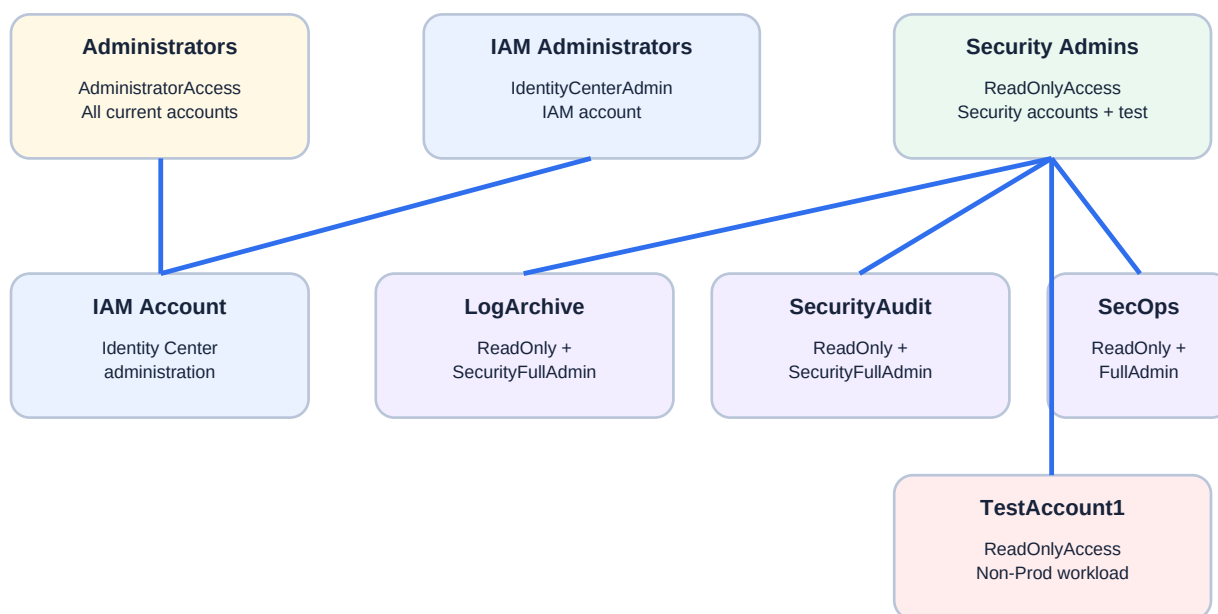
- Add the current organization accounts to the existing Administrators group with AdministratorAccess.
- Transition Identity Center administration into the delegated IAM account.
- Create Security Administrators and IAM Administrators groups.
- Create a ReadOnlyAccess permission set.
- Create an IdentityCenterAdmin permission set using the AWSSSOMemberAccountAdministrator AWS managed policy.
- Assign the SSO user to IAM Administrators with the IdentityCenterAdmin permission set.
- Assign LogArchive, SecurityOperations, SecurityAudit, and TestAccount1 access to Security Administrators using ReadOnlyAccess.
- Create a SecurityFullAdmin permission set using AdministratorAccess.
- Assign SecurityFullAdmin to LogArchive, SecurityOperations, and SecurityAudit for security response capability.

Environment

Primary Service	AWS IAM Identity Center
Delegated Admin Account	IAM
Existing Group	Administrators
New Groups	Security Administrators and IAM Administrators
Permission Set - Read	ReadOnlyAccess

Permission Set - Identity Admin	IdentityCenterAdmin using AWSSSOMemberAccountAdministrator
Permission Set - Security Full Admin	SecurityFullAdmin using AdministratorAccess
Accounts Given AdministratorAccess	IAM, LogArchive, SecurityAudit, SecurityOperations, and TestAccount1
Security Admin Read Scope	LogArchive, SecurityOperations, SecurityAudit, and TestAccount1
Security Admin Full Scope	LogArchive, SecurityOperations, and SecurityAudit
Purpose	Persona-based security administration with reduced reliance on full admin sessions

Architecture



Security personas let the same SSO user select only the access level needed for a session.

Figure 1 - IAM Identity Center persona-based access model for security and IAM administration.

Implementation Summary

1. Added IAM, SecurityAudit, SecurityOperations, and TestAccount1 to the existing Administrators group with AdministratorAccess.
2. Transitioned to the delegated IAM account to manage IAM Identity Center from the delegated administration account.
3. Created the Security Administrators group and added the SSO user.

- 4. Created the IAM Administrators group and added the SSO user.
- 5. Created the ReadOnlyAccess permission set.
- 6. Created the IdentityCenterAdmin permission set using the AWS managed policy AWSSSOMemberAccountAdministrator.
- 7. Assigned the SSO user to the IAM Administrators group with the IdentityCenterAdmin permission set.
- 8. Assigned LogArchive, SecurityOperations, and SecurityAudit access to Security Administrators using ReadOnlyAccess.
- 9. Created the SecurityFullAdmin permission set using the AdministratorAccess AWS managed policy.
- 10. Assigned SecurityFullAdmin to LogArchive, SecurityOperations, and SecurityAudit.
- 11. Added TestAccount1 to Security Administrators with ReadOnlyAccess for workload investigation visibility.

KEY CONCEPT

This project separates **identity administration**, **security read access**, and **security full-admin response access**. That allows a user to select the least-privileged persona needed for the session instead of defaulting to full administrator access.

Permission Assignment Map

Administrators	AdministratorAccess	IAM, LogArchive, SecurityAudit, SecurityOperations, TestAccount1
IAM Administrators	IdentityCenterAdmin	IAM
Security Administrators	ReadOnlyAccess	LogArchive, SecurityAudit, SecurityOperations, TestAccount1
Security Administrators	SecurityFullAdmin	LogArchive, SecurityAudit, SecurityOperations

Evidence

Figure 2 - IAM Identity Center Groups Created

☐	Group name	Description	Created by
☐	Security Administrators	-	Manual
☐	Administrators	Full AdminAccess	Manual
☐	IAM Administrators	Can manage Identities.	Manual

The IAM Identity Center groups evidence shows Security Administrators, Administrators, and IAM Administrators. IAM Administrators includes the description Can manage Identities, and Administrators retains the full admin access description.

Figure 3 - IdentityCenterAdmin Permission Set Created

AWS managed policies (1)

Policy name	Type	Description
 AWSSSOMemberAccountAdministrator	AWS managed	Provides access within AWS SSO to ma...

Customer managed policies (0)

No customer managed policies attached

Inline policy (not set)

You have not created an inline policy.

Permissions boundary (not set)

No permissions boundary set

Step 3: Define permission set details

Edit

Permission set details

Permission set name

IdentityCenterAdmin

Description

Admin AWS IAM Identity Center from Delgate Admin account.

Session duration

1 hour

Relay state

The permission-set evidence shows the AWS managed policy *AWSSSOMemberAccountAdministrator* attached to the *IdentityCenterAdmin* permission set. The details show a 1-hour session duration and the purpose of administering IAM Identity Center from a delegated admin account.

Figure 4 - Security Administrators Group Membership

Security Administrators

Delete group

▼ General Information

Group name

Security Administrators

Created time

July 15, 2026

Last updated

July 15, 2026

Group ID

Created by

Manual

Updated by

Manual

Description

Users

AWS accounts

Applications

Users in this group (1)

Remove users from group

Add users to group

Workforce users in this group inherit permissions to the AWS accounts and Identity Center enabled applications that are assigned to this group.

Find users by username or display name

Search for users in this group

< 1 > ⚙

☐

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The *Security Administrators* group evidence shows the group name, creation and update date, manual creation method, and one user in the group. Sensitive user identifiers were redacted.

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Figure 5 - Final AWS Access Portal Account and Permission Set View

Account Name	Account ID	Email
▼ IAM		
AdministratorAccess		Access keys
▼ LogArchive		
AdministratorAccess		Access keys
ReadOnlyAccess		Access keys
SecurityFullAdmin		Access keys
AdministratorAccess		Access keys
▼ SecurityAudit		
AdministratorAccess		Access keys
ReadOnlyAccess		Access keys
SecurityFullAdmin		Access keys
▼ SecurityOperations		
AdministratorAccess		Access keys
ReadOnlyAccess		Access keys
SecurityFullAdmin		Access keys
▼ TestAccount1		
AdministratorAccess		Access keys
ReadOnlyAccess		Access keys

The final access evidence shows multiple account personas available through the access portal. IAM includes AdministratorAccess; LogArchive, SecurityAudit, and SecurityOperations show AdministratorAccess, ReadOnlyAccess, and SecurityFullAdmin; TestAccount1 shows AdministratorAccess and ReadOnlyAccess.

Security Analysis

The ReadOnlyAccess persona supports investigation and assessment without requiring full administrative privileges.

The IdentityCenterAdmin permission set keeps IAM Identity Center operations in the delegated IAM account instead of using the management account for routine identity administration.

SecurityFullAdmin is intentionally powerful and should be treated as break-glass or response access. It is scoped to security accounts rather than every account.

Giving Security Administrators ReadOnlyAccess to TestAccount1 supports workload investigation while avoiding unnecessary write access in the non-production workload account.

The final access portal view demonstrates selectable personas by account. This is useful operationally because the user can choose read-only or full-admin access based on the task.

Lessons Learned

IAM Identity Center permission sets are the building blocks for role-like personas in AWS accounts.

Access assignment is a three-part relationship: group, permission set, and account.

Different security tasks require different levels of access. Read-only investigation, IAM administration, and incident response should not all use the same permission level.

Delegated IAM Identity Center administration becomes more useful once the IAM account has its own Identity Center administration permission set.

Production Improvements

- Use Just-in-Time approval for SecurityFullAdmin in production.
- Consider ViewOnlyAccess or SecurityAudit instead of ReadOnlyAccess when data exposure must be minimized.
- Create separate permission sets for security engineering, incident response, audit, and IAM operations.
- Monitor Identity Center group changes, permission-set changes, and account assignments through CloudTrail.
- Periodically review access portal assignments and remove stale account or group access.
- Use separate break-glass procedures for emergency response access.

Skills Demonstrated

IAM Identity Center	Permission sets	Group-based authorization
ReadOnlyAccess	AWSSSOMemberAccountAdministrator	AdministratorAccess
Delegated administration	Persona-based access	Least privilege

Resume Bullet

RESUME	Implemented persona-based AWS IAM Identity Center access by creating Security Administrators and IAM Administrators groups, building ReadOnlyAccess, IdentityCenterAdmin, and SecurityFullAdmin permission sets, and assigning them across security and workload accounts to reduce default full-admin usage.
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STAR Interview Story

Situation: The AWS organization had centralized SSO and delegated administration, but access was still concentrated around broad AdministratorAccess.

Task: Create security-team personas that support read-only investigation, IAM Identity Center administration, and full-admin security response without using the management account for routine work.

Action: I created Security Administrators and IAM Administrators groups, created ReadOnlyAccess, IdentityCenterAdmin, and SecurityFullAdmin permission sets, and assigned them to IAM, security, and workload accounts based on role purpose.

Result: The environment now supports least-privilege session selection through IAM Identity Center, with separate access paths for IAM administration, security read access, and security full-admin response.

Version History

v1.0.0	July 15, 2026	Initial documented implementation. Created security and IAM administrator groups, created ReadOnlyAccess, IdentityCenterAdmin, and SecurityFullAdmin permission sets, assigned personas across IAM, security, and workload accounts, and embedded provided evidence using portfolio documentation standard v1.0.1.

References

- Cloud Security Lab a Week (S.L.A.W.) - Creating Security Team Permissions in IAM Identity Center.
- User-provided IAM Identity Center groups screenshot.
- User-provided IdentityCenterAdmin permission set screenshot.
- User-provided Security Administrators group screenshot.
- User-provided final AWS access portal permission set screenshot.
- Project 20 - Delegated Administration for IAM Identity Center and CloudTrail.